

TECHNOLOGY, INNOVATION AND CREATIVITY

Technology

Technology is a body of knowledge devoted to creating tools, processing actions and the extraction of materials.

It can also be the knowledge of techniques, processes or it can be embedded in machines to allow for operation without detailed knowledge of their workings.

Technology is the collection of techniques, skills, methods and processes used in the production of goods and services or accomplishment of objectives such as scientific investigation.

It is also the application of new knowledge and techniques in production.

Types of technology

They include;

1. Information technology-it provides access to knowledge and other information resources
2. Technology productivity tools-it refers to technology that enable us to work more efficiently and effectively e.g. computers hardware and software,
3. Medical technology- advanced equipment that are used in medical field
4. Instructional technology- equipment used in the process of teaching and learning.

Innovation

This is the introduction of new techniques, product/production or distribution processes that may be followed by a spread process of diffusion.

Innovation – it is practical application of new inventions. It is the process of implementing something new.

Invention or creativity or novelty refers to the process of devising a new idea, or thing or improving an existing idea or thing.

Innovation turns new concepts into realities, creating wealth and power.

Innovation can also disrupt the status quo.for example computerization through **creative destruction**. Creative destruction occurs when innovations make long-standing arrangement obsolete and frees resources to be employed elsewhere leading to greater economic efficiency.

Types of innovation

1. **Product innovation**

This is a good or service that is new or significantly improved. This includes significant improvements in technical specifications, components and materials, software in the product, user friendliness or other functional characteristics.

2. Process innovation

This is a new or significantly improved production or delivery method. This includes significant changes in techniques, equipment and/or software.

3. Business model innovation

This reframes an established value proposition to the customers or a company's established role in the value chain or both – e.g. Gillette's move from razors to razor blades, apple move to consumer retailing etc.

4. Marketing innovation

This is a new marketing method involving significant changes in product design or packaging, product placement, product promotion or pricing. Improves customers touching processes, be they marketing processes e.g. Use of the web/e-commerce or consumer transaction, use of auction, e-commerce etc.

5. Structural innovation

Capitalizes on disruption to restructure industry relationships e.g. Deregulation/liberalization of financial sector brought many players. Family Finance, K-Rep, CFC etc. to offer broader arrays of products and services to consumers becoming major competitors to established banks – Equity

6. Entrepreneurial Innovation

The innovation dimension of entrepreneurship refers to the pursuit of creative or novel solutions to challenges confronting a firm. These challenges can include developing new products and services or new administrative techniques and technologies for performing organizational functions (e.g., production, marketing, and sales and distribution).

7. Technological Innovation

This takes place when companies try to gain a competitive advantage either by reducing costs or by introducing a new technology. Technological innovation has been a hot topic in recent years, particularly when coupled with the concept of disruptive innovation. Disruptive innovation is usually a technological advancement that renders previous products/services (or even entire industries) irrelevant. For example, the smartphone disrupted landlines, Netflix made Blockbuster obsolete, and mp3s have marginalized CD players.

Reasons for Innovation

- Innovation is essential for the entrepreneur in solving the inefficiency problems.
- It a means of cost reduction and imposing significance social and market grip.
- Profit improvements are looked at from the innovation point of view through newer technology in management and production.

- To counter competition from already established businesses.
- To facilitate opening up of new markets both locally and internationally.
- To facilitate diversification of products risks and losses.
- To protect current position of monopoly or success.

Reasons for Opposing Innovation

- The entrepreneurs tend to have a practical concern that unforeseen innovation may cause a disaster for example the side effects of a drug.
- Fear of loosing profits in the event innovation does not translate to the expectations.
- Where the entrepreneur holds a monopoly position in the market, there is fear of losing authority and control.
- Fear of upsetting the moral and social value of demand for the product.
- Desire to preserve the existing market confidence.
- Fear of upsetting tradition in production management and market scope.
- Fear of opening a loophole to competition hence lose of business grip.

THE CREATIVE PROCESS (DESIGN THINKING)

Creativity

Creativity is the process of bringing something new into being.

Creativity is the act of turning new and imaginative ideas into reality.

Design

This is a structured process that transforms creative ideas into concrete products, services and systems and as such links creativity to innovation.

Creativity is the generation of new ideas, design is the shaping (or transformation) of ideas into new products and processes and innovation is the exploitation of ideas i.e. the successful marketing of these new products and processes.

Creative process

Generation of ideas occurs within a social context and is linked to domains of knowledge and understanding that are also in a constant state of change.

Individuals require a certain level of intelligence, be willing to think in non-traditional ways and to be persistent over time.

It is argued that it is not simply the creation of new ideas that is important but the translation of these ideas into realizable products and services.

Wallas (1926) identified four stages in the creative process namely;

1. **Preparation** – refers to the period when an individual may refine their goals in response to a particular issue or question that they face.
This is also the period where relevant material from a wide range of secondary and primary sources is collected. The aim of this stage is for the individual to conduct research in order to broaden their view of the area under investigation.
2. **Incubation stage** - here individuals suspend their conscious concentration on the problem and engage in a process of subconscious data processing.
3. **Illumination stage** – this is the period when someone suddenly becomes aware of a core answer to the problem. It is characterized by an unplanned result that derives from a unique combination of ideas or patterns of knowledge that occurs during the incubation stage. It is during this period that creative individuals must use their logical thought processes to turn sudden insight into a novel and valuable solution.
4. **Verification stage** - this is the transition of new ideas into a realizable solution. It is when the individual needs to formally evaluate the resultant outcome against the criteria set at the outset.

Basadur's et al., (1982) three stage model of process of creative problem solving

1. **Problem finding**
They argue that in order for creative process to begin there needs to be a problem that requires a solution. This stage is important because the way one approaches a problem will affect the quantity and quality of ideas generated in the stages that follow.
2. **Problem solving**
Focuses on generating as many ideas as possible, it is assumed that the number of ideas increases the probability of someone coming up with an idea worth pursuing.
3. **Solution implementation**
This focuses on implementation of one of the solutions that were generated from the previous stage.

NB: During each of the three stages a two-step process of ideation-evaluation occurs. Ideation refers to the uncritical development of ideas. Evaluation refers to the selection of the best of the generated ideas.

Amabile (1983) five stage- componential model

This model identifies key components of creativity at certain stages of the creative process.

1. **Problem or task presentation**
The task to be undertaken or the problem to be solved is presented to the creative person. It can arise from external stimuli (the supervisor may have assigned the task) or from internal stimuli (one may particularly be interested in solving a specific problem).

2. Preparation

At this stage the creative employee develops or reactivates a store of data relevant to the problem or the opportunity identified.

3. Response generation

Here the individual comes up with a diverse range of possible ideas appropriate to the issue in question. It is at this stage that an individual's creative thinking will determine both the quality and quantity of ideas generated.

4. Response validation

This refers to the process through which new ideas are checked for their appropriateness and validated.

5. Outcome assessment based on tests performed in the previous stage.

If the response is found to be wholly appropriate then the outcome will be accepted and the process ends. If, however, the response is unacceptable or only partially acceptable but shows potential then the process returns to the initial stage of problem or task presentation.

Design thinking process

Design thinking is a cognitive process from which ideas of a product emerge. It is related to problem solving, decision making, sketching and developing a prototype. Design thinking is the method of developing designs as a mechanism to provide solutions to existing problems. It is a human centered creative approach that focuses on the needs of the end user.

The design thinking concept involves a series of steps that entrepreneurs engage in when developing products and services that address the needs, pains, fears of the customer. The design thinking process involves five steps: Empathize, Define, Ideate, Prototype, and Test. The design thinking process involves a series of steps that utilize the aspects of imagination, intuition, logic, creativity, and systematic reasoning to bring out products or services that provide the benefits that satisfy the end user.

Stages of thinking

Design is a process that turns a brief or requirement into a finished product or design solution.

The design process can be said to comprise seven stages: **define, research, ideate, prototype, select, implement and learn.**

The design process engages a high degree of creativity but in a way that is controlled and directed by the process so that it is channelled towards producing a viable, practical solution to the design problem, meeting or excelling the stated aims of the brief.

While creativity in design is important, design is an activity that serves economic as well as creative goals. The design process helps ensure that a design satisfies all such considerations. The process seeks to generate a number of possible solutions and utilises various techniques or mechanisms that encourage participants to think outside the box in the pursuit of creative or innovative solutions.

The design process

Within the design process, seven steps can be identified: define, research, ideate, prototype, select, implement and learn.

First, the design problem and the target audience need to be **defined**. A precise understanding of the problem and its constraints allows more exact solutions to be developed. This stage determines what is necessary for the project to be successful.

The **research** stage reviews information such as the history of the design problem, end-user research and opinion-led interviews, and identifies potential obstacles.

Ideate is the stage where end-user motivations and needs are identified and ideas are generated to meet these, perhaps through brainstorming.

Prototyping sees the resolve or working-up of these ideas, which are presented for user-group and stakeholder review, prior to being presented to the client.

Selection sees the proposed solutions reviewed against the design brief objective. Some solutions might be practical but may not be the best ones.

Implementation sees design development and its final delivery to the client.

Learning helps designers improve their performance and, for this reason, designers should seek client and target audience feedback and determine if the solution met the goals of the brief. This may identify improvements that can be made in the future.

Value proposition model by Peter Thompson

Peter Thompson proposed that entrepreneurs can use the pains and gains value proposition model in the design thinking process to develop a product that can minimize the customers pains while creating some gains so as to satisfy the customer. His proposition aids the entrepreneur to help the

customer take care of their needs and wants by producing a product that has benefits to this customer



The concept of design thinking is a user centered approach in which the entrepreneur immerses into the lives of the customers to identify the pains and problems that they could be undergoing and finding solutions to those problems. Hence the design thinking process is a problem-solution exploration process. Once the entrepreneur has developed the value proposition model discussed earlier (that explains the pains, gains, and jobs to be done by the customer in addition to the benefits provided by the products and services) the next step is to develop a business model canvas that allows him/her to organize the business plan in a logical way.

Business Model Canvas

The **Business Model Canvas** is a template used for developing new business models and documenting existing ones. It offers a visual chart with elements describing a firm's or product's value proposition, infrastructure, customers, and finances, assisting businesses to align their activities by illustrating potential trade-offs.

The nine "building blocks" of the business model design template that came to be called the Business Model Canvas were initially proposed in 2005 by Alexander Osterwalder, based on

his earlier work on business model ontology. Since the release of Osterwalder's work around 2008, new canvases for specific niches have appeared.

The Business Model Canvas

Designed for:

Designed by:

Date:

Version:

Key Partners Who are our key partners? Who are our key suppliers? Which key resources do we acquire from partners? Which key activities do partners perform? Key Partnerships Do partners help us reduce costs? Do partners help us increase revenue? Do partners help us create new value?	Key Activities What key activities do we do to create value? What key activities do we do to create value? What key activities do we do to create value? Key Resources What key resources do we need to create value? What key resources do we need to create value? What key resources do we need to create value?	Value Propositions What value do we deliver to the customer? What value do we deliver to the customer? What value do we deliver to the customer? Channels Through which channels do we reach our customers? Through which channels do we reach our customers? Through which channels do we reach our customers?	Customer Relationships What type of relationship does each of our customer segments expect us to establish and maintain with them? What type of relationship does each of our customer segments expect us to establish and maintain with them? Customer Segments For whom are we creating value? Who are our most important customers? Who are our most important customers?
Cost Structure What are the most important costs we incur to create value? What are the most important costs we incur to create value? What are the most important costs we incur to create value?		Revenue Streams For what value are our customers really willing to pay? For what value are our customers really willing to pay? For what value are our customers really willing to pay?	

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Osterwalder's canvas has nine boxes: customer segments, value propositions, channels, customer relationships, revenue streams, key resources, key activities, key partnerships, and cost structure.

- Key activities:** The most important activities in executing a company's value proposition. An example for Bic, the pen manufacturer, would be creating an efficient supply chain to drive down costs.
- Key resources:** The resources that are necessary to create value for the customer. They are considered assets to a company that are needed to sustain and support the business. These resources could be human, financial, physical and intellectual.
- Partner network:** In order to optimize operations and reduce risks of a business model, organizations usually cultivate buyer-supplier relationships so they can focus on their core activity. Complementary business alliances also can be considered through joint ventures or strategic alliances between competitors or non-competitors.
- Value propositions:** The collection of products and services a business offers to meet the needs of its customers. According to Osterwalder (2004), a company's value

proposition is what distinguishes it from its competitors. The value proposition provides value through various elements such as newness, performance, customization, "getting the job done", design, brand/status, price, cost reduction, risk reduction, accessibility, and convenience/usability.

- The value propositions may be:
 - Quantitative – price and efficiency
 - Qualitative – overall customer experience and outcome
- 5. **Customer Segments:** To build an effective business model, a company must identify which customers it tries to serve. Various sets of customers can be segmented based on their different needs and attributes to ensure appropriate implementation of corporate strategy to meet the characteristics of selected groups of clients. The different types of customer segments include:
 - Mass market: There is no specific segmentation for a company that follows the mass market element as the organization displays a wide view of potential clients: e.g. car.
 - Niche market: Customer segmentation based on specialized needs and characteristics of its clients: e.g. Rolex.
 - Segmented: A company applies additional segmentation within existing customer segment. In the segmented situation, the business may further distinguish its clients based on gender, age, and/or income.
 - Diversify: A business serves multiple customer segments with different needs and characteristics.
 - Multi-sided platform/market: For a smooth day-to-day business operation, some companies will serve mutually dependent customer segments. A credit card company will provide services to credit card holders while simultaneously assisting merchants who accept those credit cards.
- 6. **Channels:** A company can deliver its value proposition to its targeted customers through different channels. Effective channels will distribute a company's value proposition in ways that are fast, efficient and cost-effective. An organization can reach its clients through its own channels (store front), partner channels (major distributors), or a combination of both.
- 7. **Customer Relationships:** To ensure the survival and success of any businesses, companies must identify the type of relationship they want to create with their customer segments. That element should address three critical steps on a customers relationship: How the business will get new customers, how the business will keep customers purchasing or using its services and how the business will grow its revenue from its current customers. Various forms of customer relationships include:
 - Personal assistance: Assistance in a form of employee-customer interaction. Such assistance is performed during sales and/or after sales.
 - Dedicated personal assistance: The most intimate and hands-on personal assistance in which a sales representative is assigned to handle all the needs and questions of a special set of clients.
 - Self service: The type of relationship that translates from the indirect interaction between the company and the clients. Here, an organization provides the tools needed for the customers to serve themselves easily and effectively.
 - Automated services: A system similar to self-service but more personalized as it has the ability to identify individual customers and their preferences. An example of this

would be Amazon.com making book suggestions based on the characteristics of previous book purchases.

- Communities: Creating a community allows for direct interactions among different clients and the company. The community platform produces a scenario where knowledge can be shared and problems are solved between different clients
- Co-creation: A personal relationship is created through the customer's direct input to the final outcome of the company's products/services.

8. Cost structure: This describes the most important monetary consequences while operating under different business models.

- Classes of business structures:
 - Cost-driven – This business model focuses on minimizing all costs and having no frills: e.g. low-cost airlines.
 - Value-driven – Less concerned with cost, this business model focuses on creating value for products and services: e.g. Louis Vuitton, Rolex.
- Characteristics of cost structures:
 - Fixed costs – Costs are unchanged across different applications: e.g. salary, rent.
 - Variable costs – Costs vary depending on the amount of production of goods or services: e.g. music festivals.
 - Economies of scale – Costs go down as the amount of goods are ordered or produced.
 - Economies of scope – Costs go down due to incorporating other businesses which have a direct relation to the original product.

9. Revenue streams: The way a company makes income from each customer segment.

Several ways to generate a revenue stream:

- Asset sale – (the most common type) Selling ownership rights to a physical good: e.g. retail corporations.
- Usage fee – Money generated from the use of a particular service: e.g. UPS.
- Subscription fees – Revenue generated by selling access to a continuous service: e.g. Netflix.
- Lending/leasing/renting – Giving exclusive right to an asset for a particular period of time: e.g. leasing a car.
- Licensing – Revenue generated from charging for the use of a protected intellectual property.
- Brokerage fees – Revenue generated from an intermediate service between 2 parties: e.g. broker selling a house for commission.
- Advertising – Revenue generated from charging fees for product advertising.